UNIVERSITY OF ALBERTA

November 5, 2026

Pasko Rakic, MD, PhD
Editor-in-Chief, *Cerebral Cortex*

Re: Revised Manuscript Submission — CerCor-2022-00354
"Connecting Covert Attention and Visual Perception to the Spatiotemporal Dynamics of Alpha Band Activity, Cross-Frequency Coupling (CFC), and Functional Connectivity using Multivariate Pattern Analysis (MVPA)"

Dear Dr. Rakic,

Please find enclosed our revised manuscript (**CerCor-2022-00354**), entitled *"Connecting Covert Attention and Visual Perception to the Spatiotemporal Dynamics of Alpha Band Activity, Cross-Frequency Coupling (CFC), and Functional Connectivity using Multivariate Pattern Analysis (MVPA),"* by Sarah S. Sheldon, Alona Fyshe, and Kyle E. Mathewson, for reconsideration as an Original Research Report in *Cerebral Cortex*.

We would like to thank you and the two anonymous reviewers for your constructive feedback and for the opportunity to submit a revised version of our work. Our manuscript extends current knowledge about the spatiotemporal dynamics of pre-stimulus alpha (8–14 Hz) oscillatory activity and its role in visual perception and visuospatial attention. Using single-trial electroencephalography (EEG) recorded from $N = 28$ participants performing a cued continuous orientation perception task, we compared the pre-stimulus spatiotemporal dynamics of local alpha amplitude, five local alpha–gamma cross-frequency coupling (CFC) estimators, and four inter-electrode alpha functional connectivity estimators using linear support vector machine (SVM) classification and linear support vector regression (SVR). We found that pre-stimulus alpha amplitude most consistently distinguished trials with covert spatial attention from trials without covert spatial attention across the 1000 ms cue–target interval, whereas none of the metrics reliably predicted trial-by-trial behavioral orientation reproduction errors—providing direct multivariate support for the baseline sensory excitability model (BSEM; Samaha et al. 2020).

In accordance with the reviewers' comments and your editorial instructions, we have thoroughly revised the manuscript and figures:

1. **Clarification of Study Novelty, SNR Considerations, Task Demands, and Statistical Error Control (Reviewer 1):** We expanded the Discussion to explicitly articulate what our head-to-head multivariate benchmark of 10 pre-stimulus oscillatory metrics adds to the literature, how differences in single-trial signal-to-noise ratio (SNR) between first-order amplitude and higher-order CFC/connectivity estimators affect decoding sensitivity, how continuous mouse-driven orientation reproduction introduces working-memory and visuomotor variance compared with binary detection/2-AFC tasks, why right-handed participants exhibit a visual field asymmetry in orientation error magnitude, and how within-analysis $t_{\max}$ permutation testing interacts with experiment-wise Type I error control across metrics.

2. **New MVPA Summary Comparison Figure (Figure 7) and Enhanced Figure Readability (Re-**

viewer 2): As suggested by Reviewer 2, we added a new three-panel summary figure (**Figure 7**) directly comparing peak SVM classification accuracy, temporal extent of significant decoding, and peak SVR error prediction across all 10 EEG metrics. We also divided **Figure 2** into explicit **A** and **B** panels, aligned the text order of absolute and signed response errors with Figure 2A–B, and configured our automated figure pipeline to enforce legible axis typography, concise abbreviations (`MI`, `Accuracy`, r_z), and 200 ms time-tick spacing across Figures 3–6.

3. **Methodological and Electrophysiological Precision (Reviewer 2):** We revised the Abstract, restructured the Introduction to clearly separate background literature from our hypotheses, clarified all methodological queries (post-mask fixation display, self-paced breaks, signed vs. absolute response error definitions, $F_{\max}$ permutation rationale, HEOG saccade rejection criteria, FOOOF $1/f$ vs. peak decomposition, individual alpha peak frequency summary statistics [$M = 10.6\,\text{Hz}, \text{SD} = 0.5\,\text{Hz}$], local CFC scope and sideband-preserving gamma filter bandwidths), and corrected the Discussion of invasive ECoG (Voytek 2010) versus non-invasive MEG (Bonnefond and Jensen 2015) high-gamma recordings.

4. **Journal Formatting Compliance:** All new and revised sections are clearly highlighted in the manuscript, all tables and figures with captions are embedded directly at their relevant locations within a single double-spaced document, and all references have been formatted strictly to *Cerebral Cortex* style.

A detailed point-by-point response to all reviewer comments is provided in the accompanying *Response to Reviewers* document. All raw data (https://osf.io/u6fgm/) and end-to-end reproducible task, analysis, and manuscript generation scripts (https://github.com/APPLabUofA/CuedOrientTask_paper, https://github.com/APPLabUofA/MVPA_CuedOrientTask_paper, and https://github.com/APPLabUofA/OrientExoML) remain openly available.

Thank you very much for your time and consideration.

Sincerely,

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